

Vector Equations

Geometric Algorithms
Lecture 4

Practice Problem

Suppose that A is a 322×245 augmented matrix for a system with infinitely many solutions. What is the maximum number of pivot positions that A can have?

Answer

Outline

- » Formally define vectors
- » Discuss vector operations and vector algebra
- » Draw the connection between vectors and systems of linear equations

Keywords

vector

vector addition

vector scaling/multiplication

the zero vector

vector equations

linear combinations

span

A Bit of Motivation

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Data are ordered collections of values

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Q: When are two data *similar*?

A Bit of Motivation

Data are ordered collections of values

Q: When are two data *similar*?

A: When they're **close together** geometrically

A Note on Algebra

$$\mathbf{v} = \mathbf{w}$$

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

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And object then *implements* an interface

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In programming, an *interface* is an abstract collection of related functions

$$\mathbf{v} = \mathbf{w}$$

And object then *implements* an interface

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Doing abstract algebra is like implementing an *algebraic* interface

Vectors

What is a vector (in $\mathbb{R}^n$)?

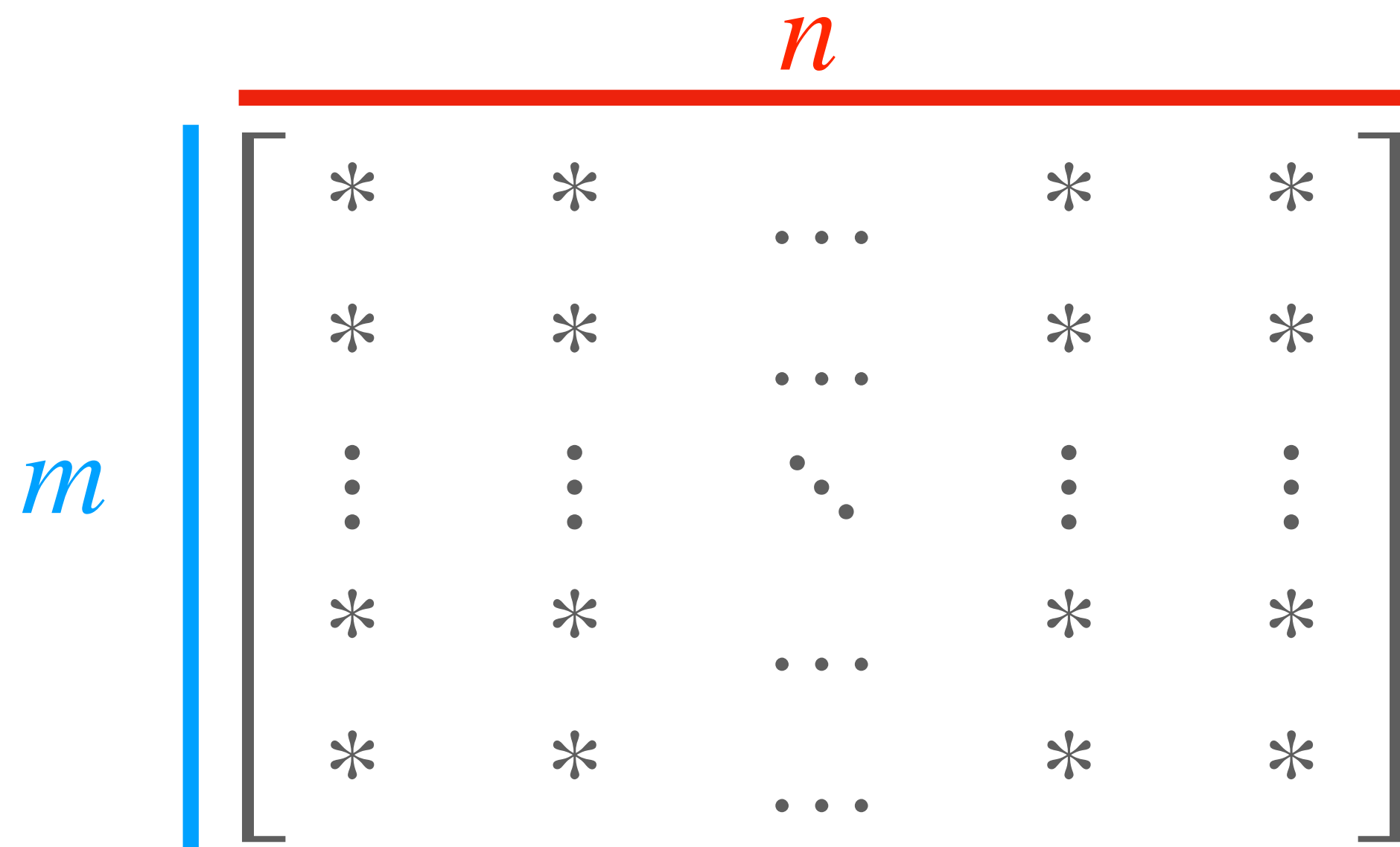
- A. an n -tuple of real numbers
- B. a point in $\mathbb{R}^n$
- C. a 1-column matrix with real values
- D. all of the above
- E. none of the above?

Column Vectors

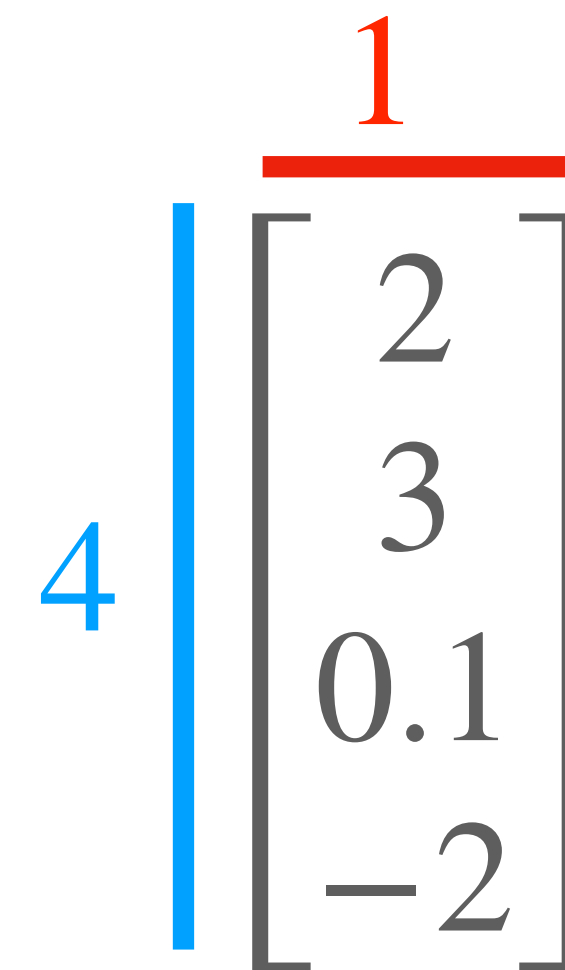
Df. a *column vector* is a matrix with a single column

Matrix Size

An $(m \times n)$ matrix has m rows and n columns



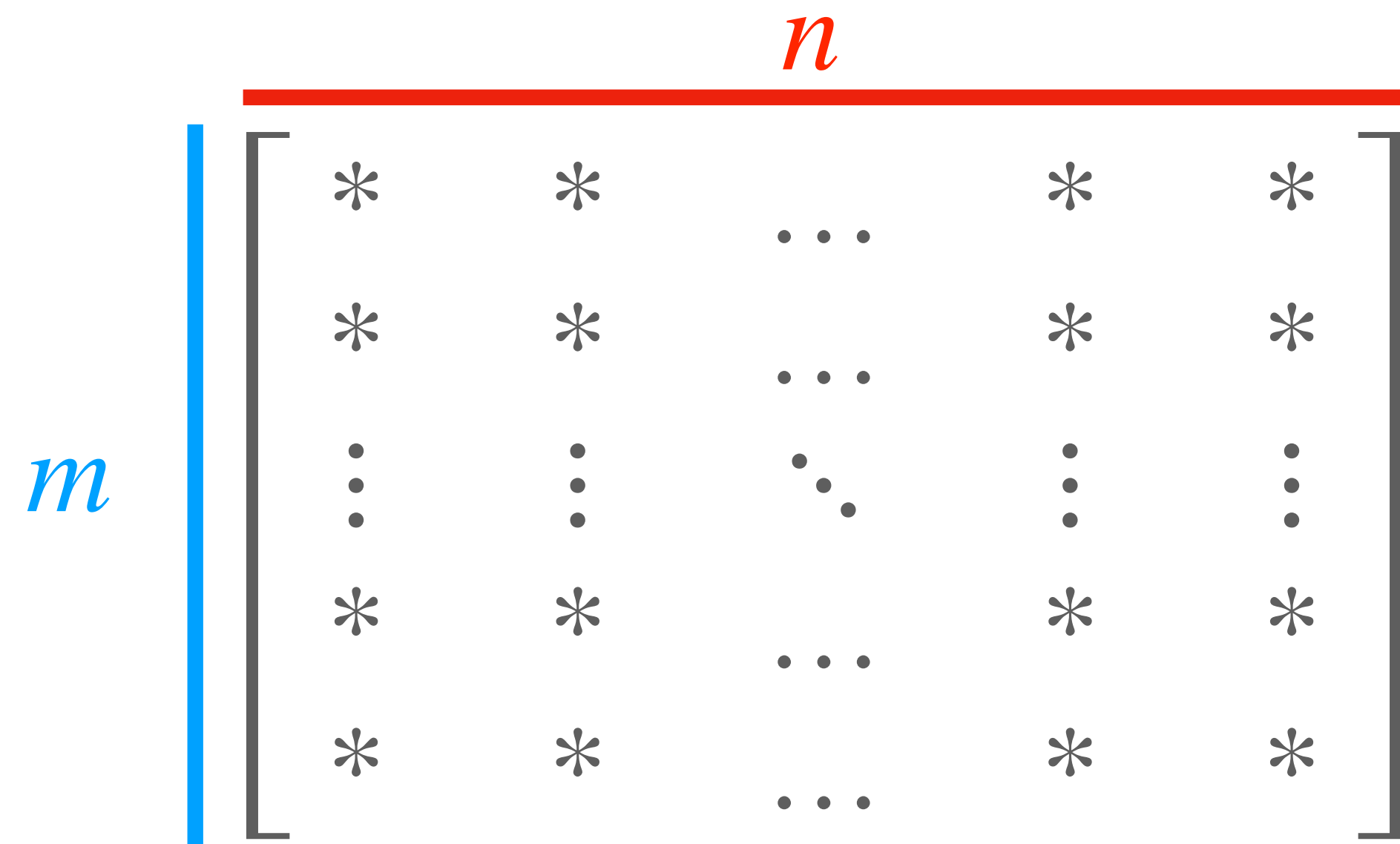
A diagram of a general matrix with m rows and n columns. A blue vertical line to the left of the matrix is labeled with the variable m in blue. A red horizontal line above the matrix is labeled with the variable n in red. The matrix is enclosed in large square brackets and contains asterisks (*) and ellipses (...) to represent its elements.



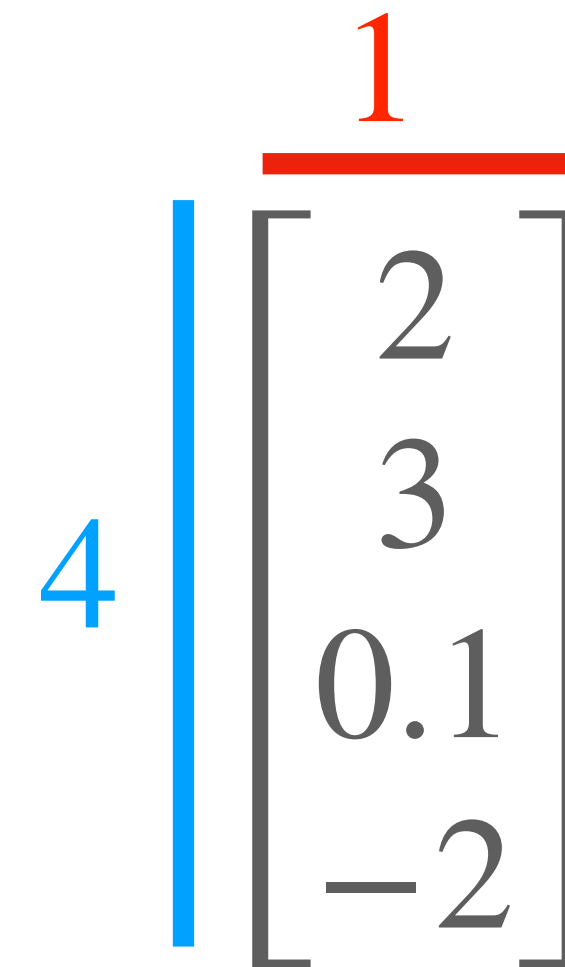
A diagram of a column vector with 4 rows and 1 column. A blue vertical line to the left of the vector is labeled with the number 4 in blue. A red horizontal line above the vector is labeled with the number 1 in red. The vector is enclosed in large square brackets and contains the numerical values 2, 3, 0.1, and -2.

Matrix Size

An $(m \times n)$ matrix has m rows and n columns



A diagram of a general matrix with m rows and n columns. A blue vertical line to the left of the matrix is labeled m . A red horizontal line above the matrix is labeled n . The matrix is enclosed in large square brackets and contains asterisks (*) and ellipses (...) to represent its entries.

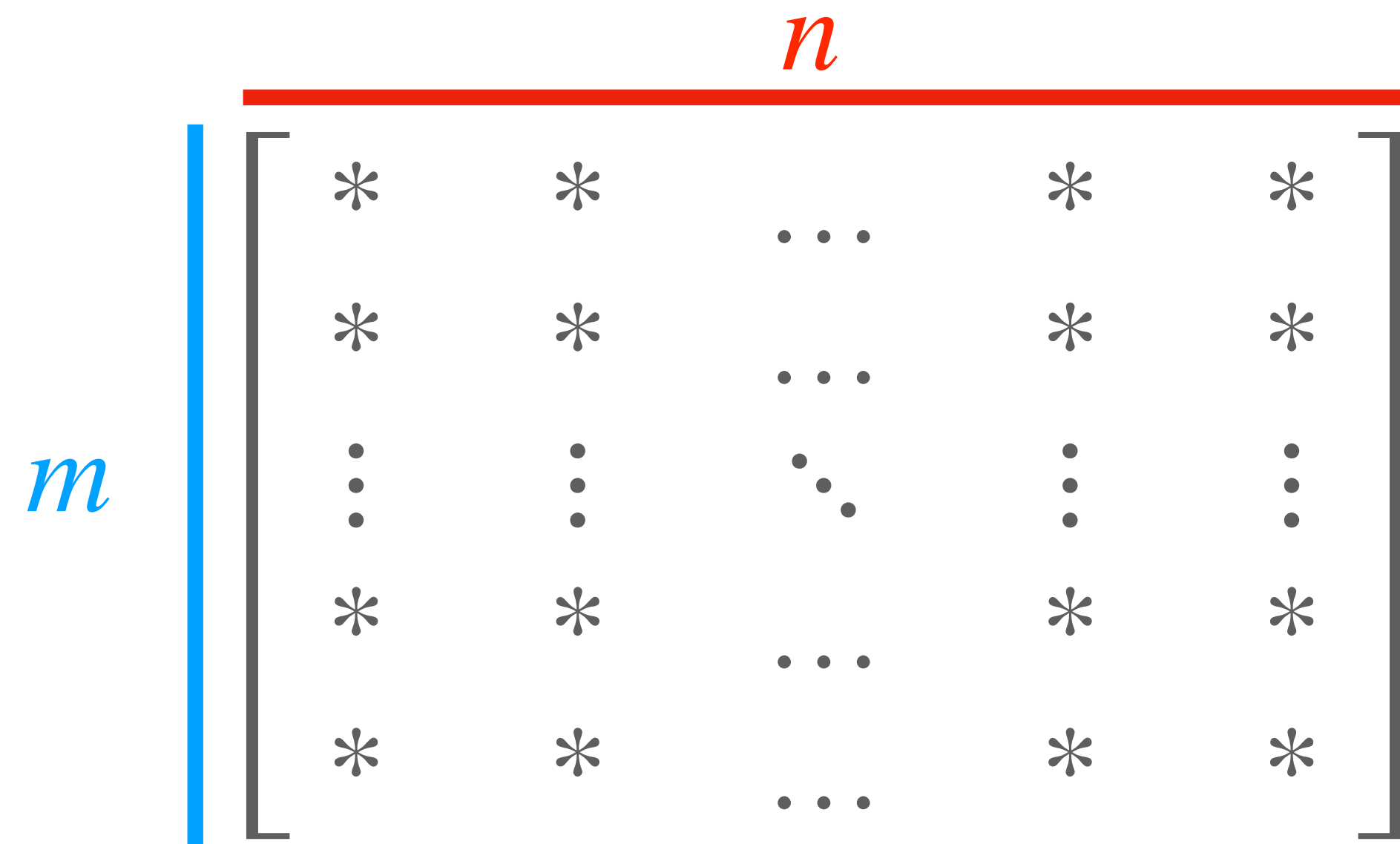


A diagram of a specific column vector with 4 rows and 1 column. A blue vertical line to the left of the vector is labeled 4. A red horizontal line above the vector is labeled 1. The vector is enclosed in large square brackets and contains the numerical values 2, 3, 0.1, and -2.

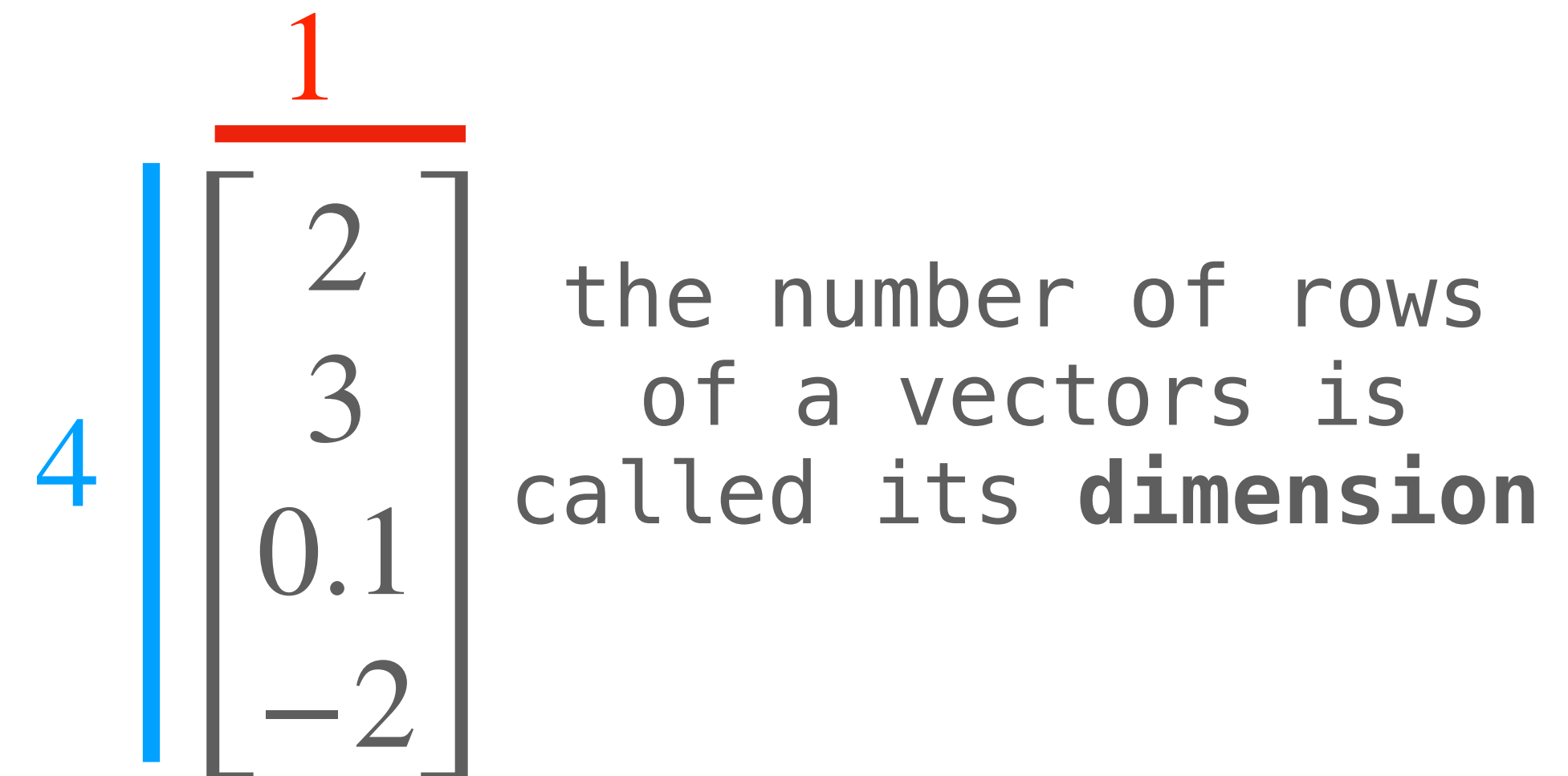
$\mathbb{R}^{m \times n}$ is set of matrices with $\mathbb{R}$ entries

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A diagram of a general matrix with m rows and n columns. A blue vertical line to the left of the matrix is labeled with a blue italic m . A red horizontal line above the matrix is labeled with a red italic n . The matrix is enclosed in large square brackets and contains asterisks (*) and ellipses (...) representing its entries.

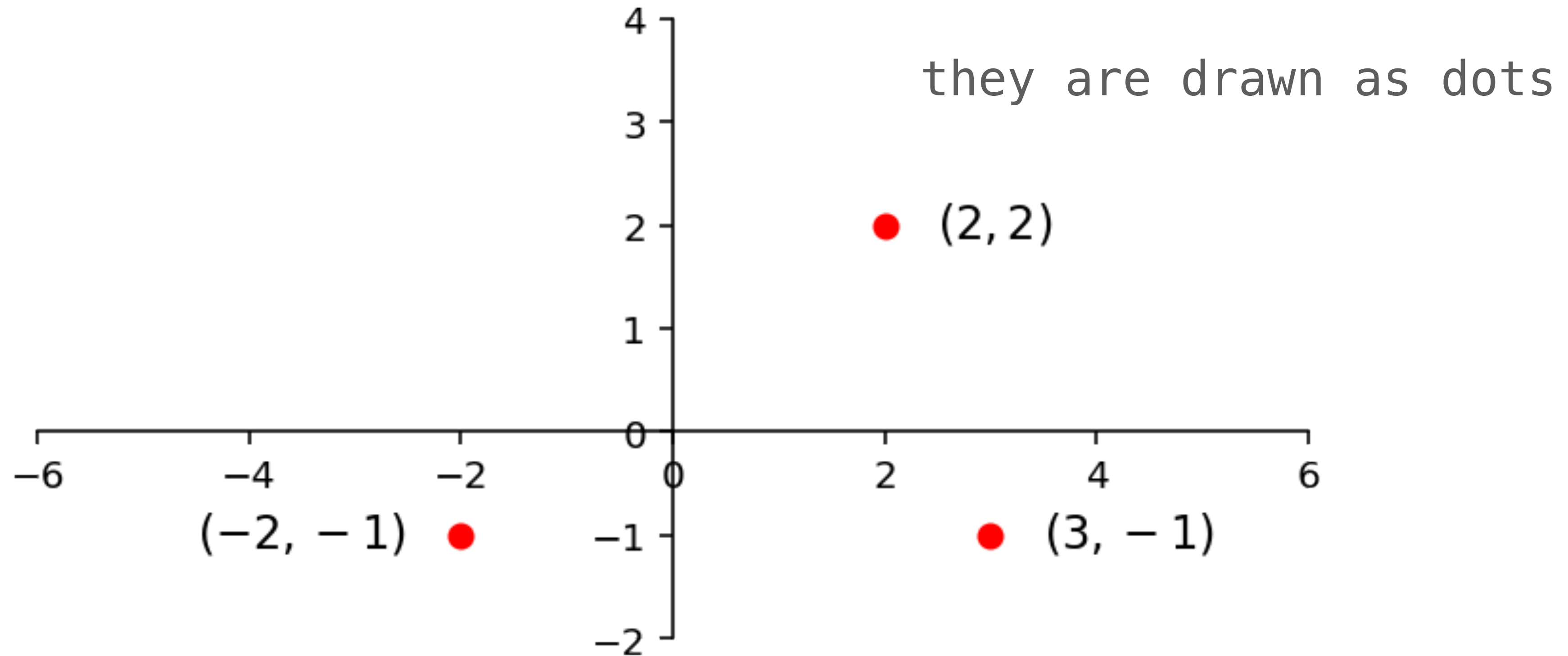


A diagram of a vector with 4 rows and 1 column. A blue vertical line to the left of the vector is labeled with a blue number 4. A red horizontal line above the vector is labeled with a red number 1. The vector is enclosed in large square brackets and contains the values 2, 3, 0.1, and -2. To the right of the vector, text explains that the number of rows of a vector is called its dimension.

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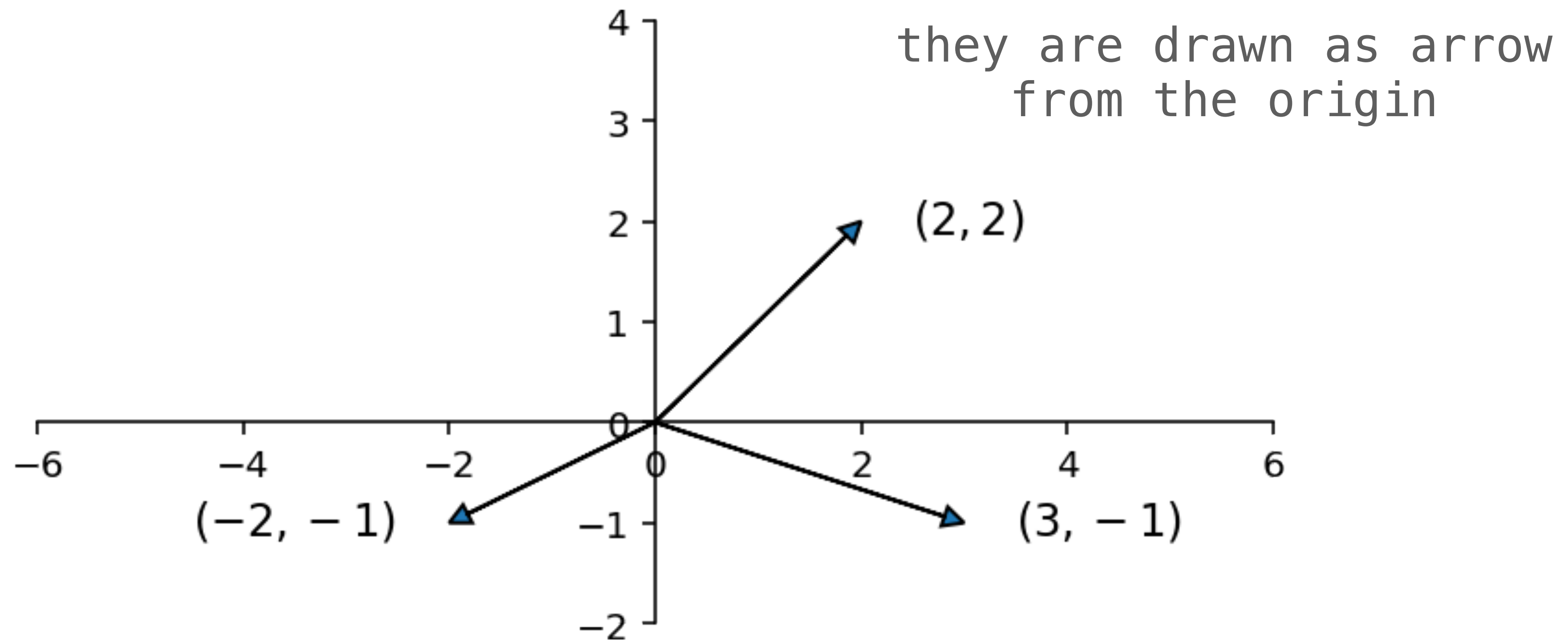
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Notation (Points)



points in $\mathbb{R}^2$ are notated as (a, b)

Notation (Vectors)



vectors in $\mathbb{R}^2$ are notated as $\begin{bmatrix} a \\ b \end{bmatrix}$

Notation (Looking ahead)

We will often write $[a_1 \ a_2 \ \dots \ a_n]^T$ for the vector $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

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IMPORTANT

$(a_1, a_2, \dots, a_n)$ is not the same as $[a_1 \ a_2 \ \dots \ a_n]$

Vector Operations

Vector "Interface"

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equality what does it mean for two vectors
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Vector Equality

Vector Equality

two vectors are equal if their entries at each position are equal

(this is also the case for matrices)

!!IMPORTANT!!
ORDER MATTERS

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Vector Equality

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \text{is the same as} \quad \begin{array}{l} a_1 = b_1 \\ a_2 = b_2 \\ \vdots \\ a_n = b_n \end{array}$$

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Vector Addition

Vector Addition

adding two vectors means adding their entries

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

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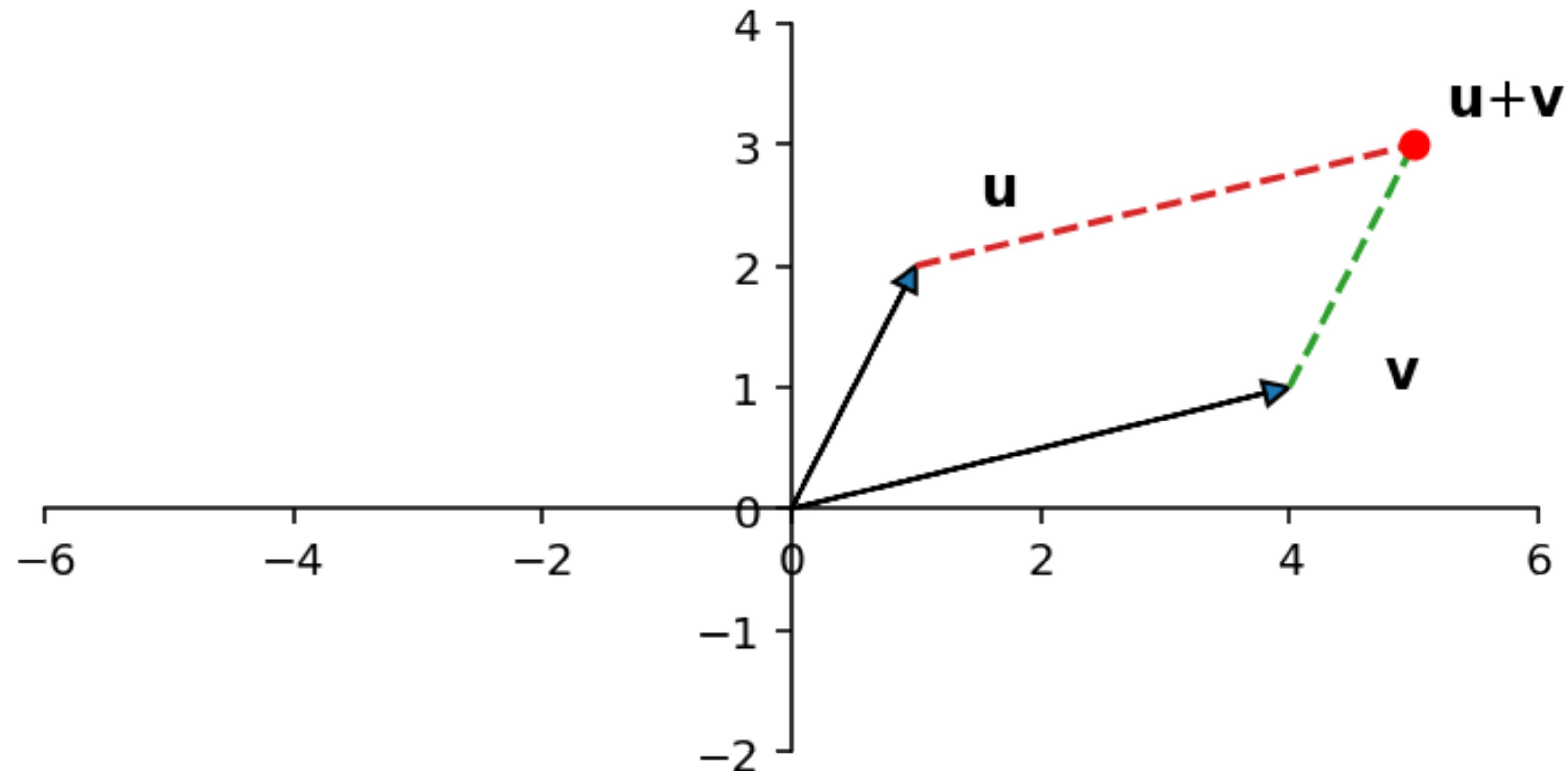
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!! IMPORTANT !!

WE CAN ONLY ADD VECTORS OF THE SAME SIZE

Vector Addition (Geometrically)

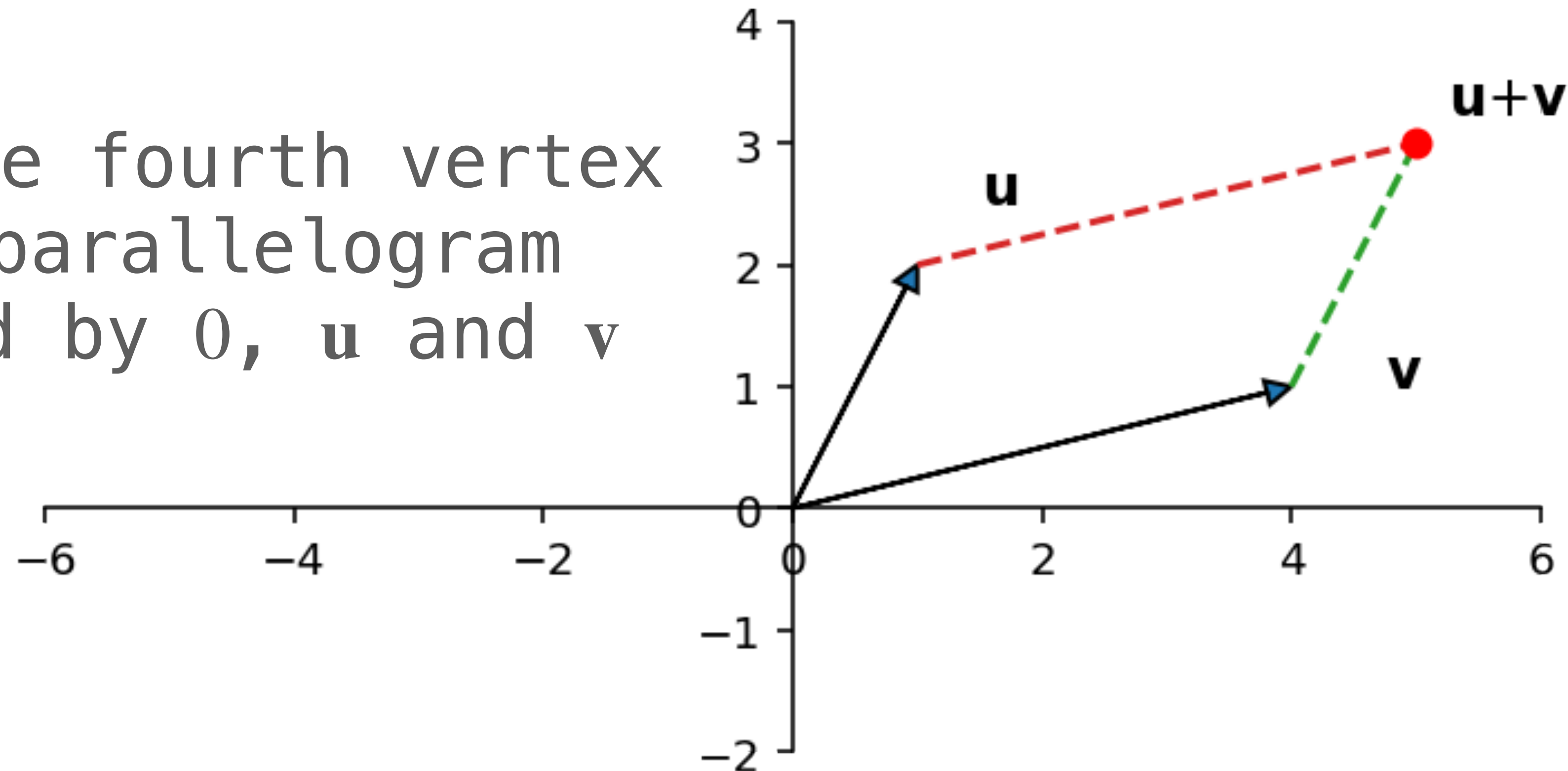
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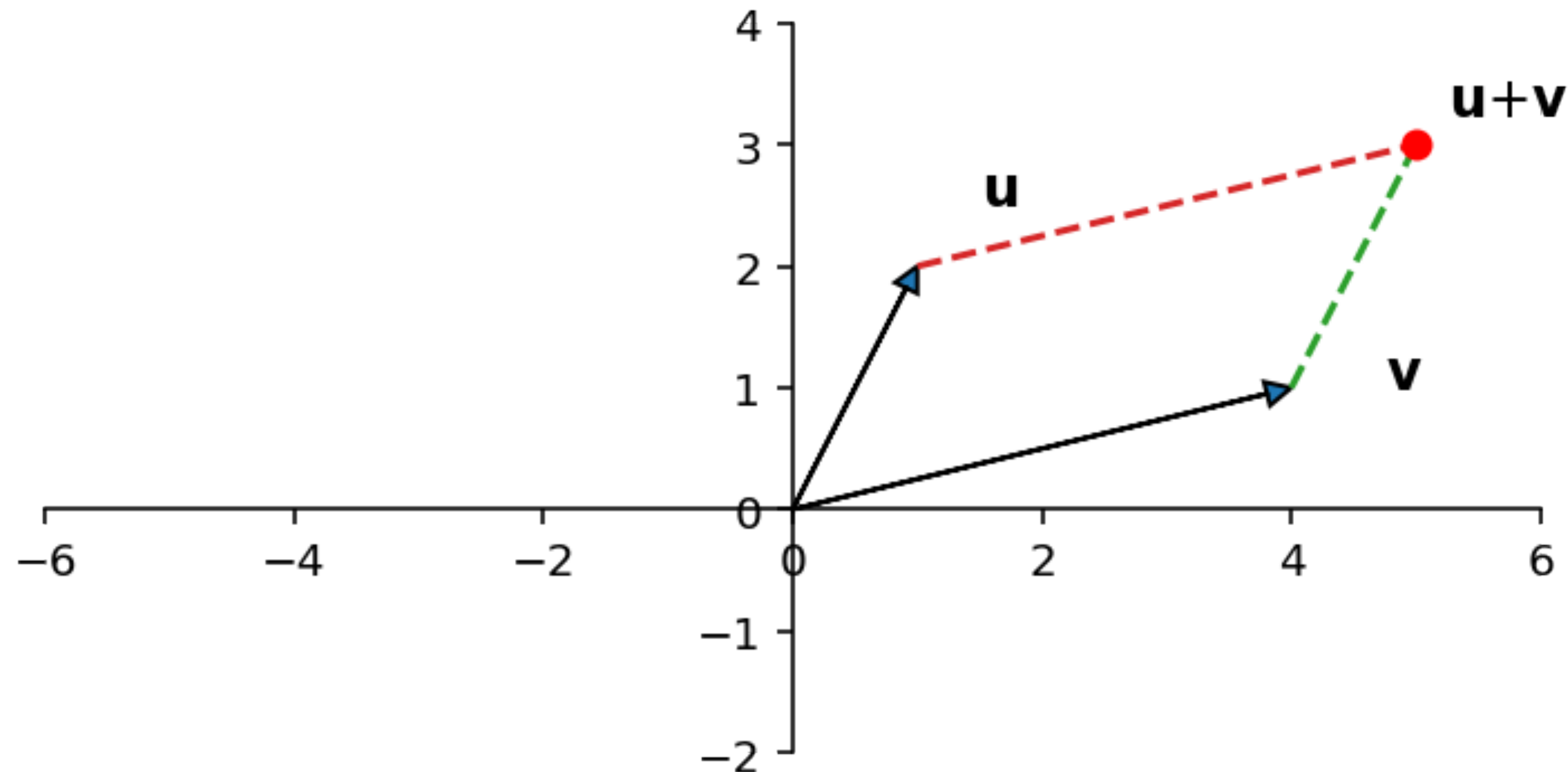
in $\mathbb{R}^2$ it's called the *parallelogram rule*

$\mathbf{u} + \mathbf{v}$ is the fourth vertex
of the parallelogram
generated by $\mathbf{0}$, $\mathbf{u}$ and $\mathbf{v}$



Vector Addition (Geometrically)

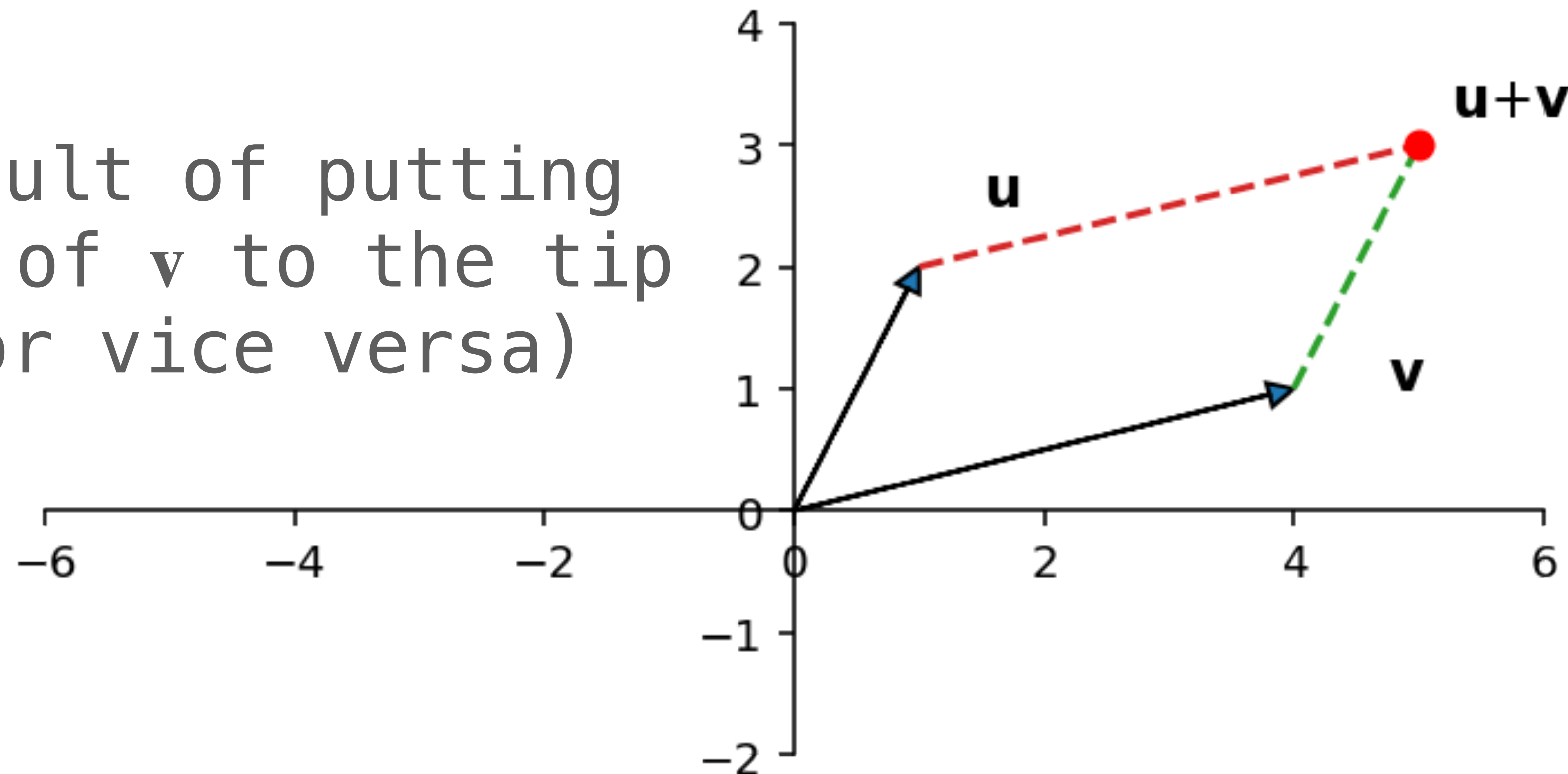
or the *tip-to-tail rule*



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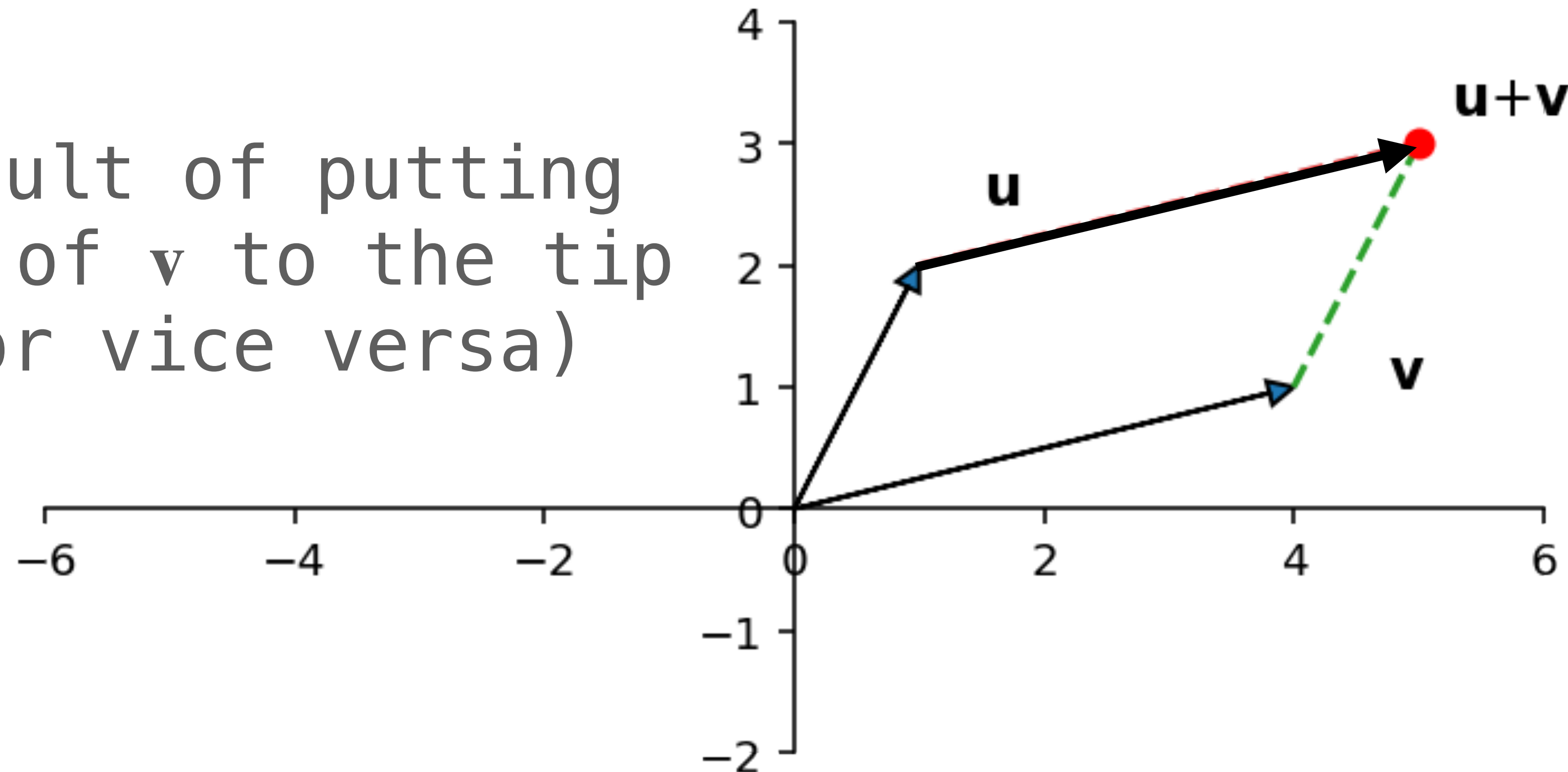
$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
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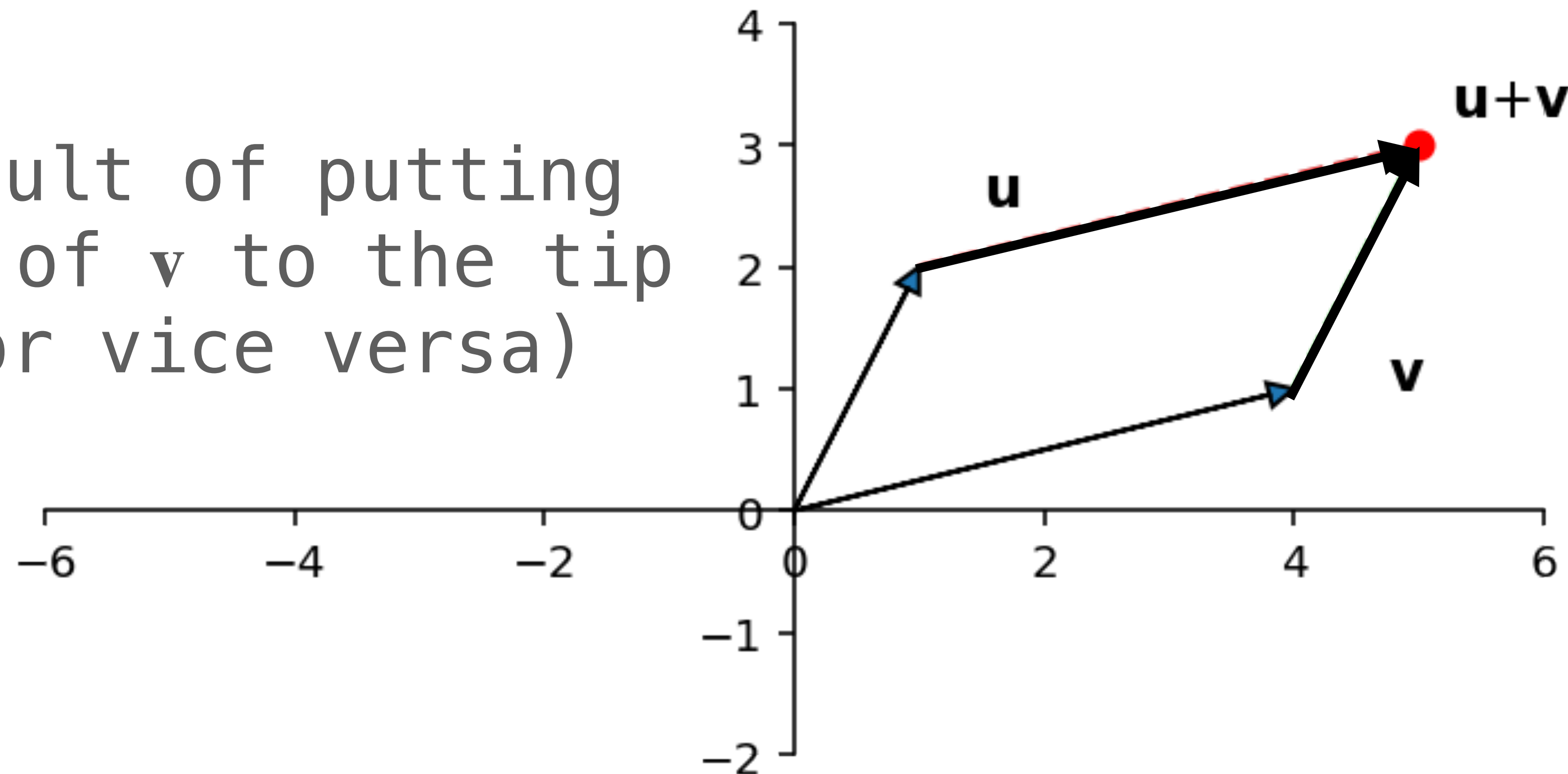
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demo
(from ILA)

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What properties do they need to satisfy?

Vector Scaling/Multiplication

Vector Scaling/Multiplication

scaling/multiplying a vector by a number means multiplying each of it's elements

$$a \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} ab_1 \\ ab_2 \\ \vdots \\ ab_n \end{bmatrix}$$

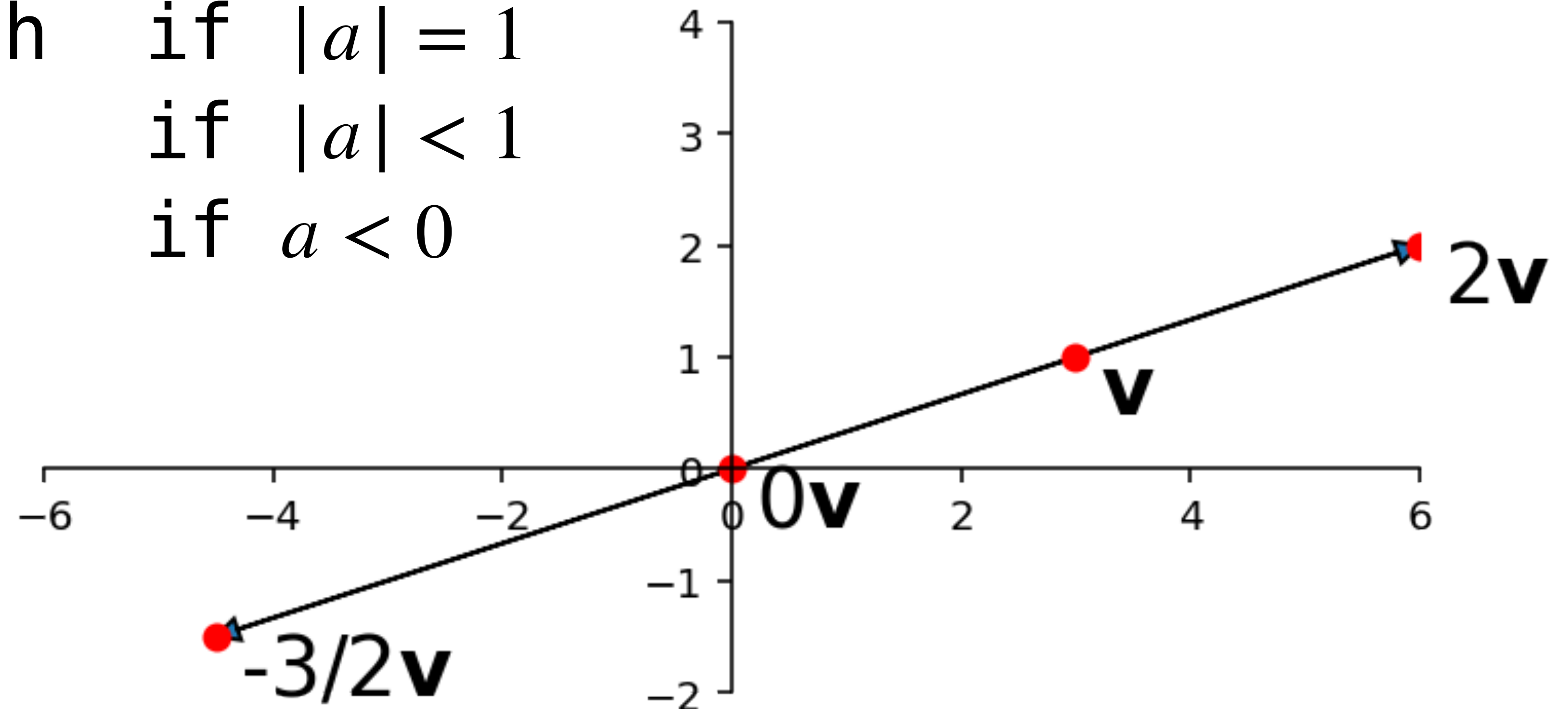
Vector Scaling/Multiplication (Example)

Scaling/multiplying a vector by a number means multiplying each of it's elements

$$3 \begin{bmatrix} 2 \\ 1 \\ 3.5 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 \\ 3 \cdot 1 \\ 3 \cdot 3.5 \\ 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 10.5 \\ 12 \end{bmatrix}$$

Vector Scaling (Geometrically)

longer	if $ a > 1$
the same length	if $ a = 1$
shorter	if $ a < 1$
reversed	if $a < 0$



demo
(from ILA)

Algebraic Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any real numbers c, d :

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$$

$$1\mathbf{u} = \mathbf{u}$$

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these are requirements for any **vector space**
they matter more for *bizarre* vector spaces

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

Example "Proof"

Question (Practice)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Compute the value of the above vector

Answer

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

In Sum

In Sum

we can add vectors

In Sum

we can add vectors

we can scale vectors

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What vectors can we make with these operations?

Linear Combinations

Linear Combinations

Linear Combinations

Df. a *linear combination* of vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$$

is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

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weights

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Looks suspiciously like
a linear equation...

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$
weights

Linear Combinations (Example)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Definition. a *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

demo
(from ILA)

The Fundamental Concern

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Can $\mathbf{u}$ be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$?

That is, are there weights $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \mathbf{u}$?

Why is this fundamental?

I'm going to ask that you suspend your disbelief...

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For now, how do we solve this problem?

Vector Equations and Linear Systems

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

The Fundamental Connection

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We want the weights...

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We want the weights...

What if we write them as unknowns?

Some Symbol Pushing...

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

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$$\begin{bmatrix} x_1 \\ (-2)x_1 \\ (-5)x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$\begin{bmatrix} x_1 + 2x_2 \\ (-2)x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

$$(-2)x_1 + 5x_2 = 4$$

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Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

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we get a system
of linear
equations we
know how to
solve

General Vector Equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 \\ a_{21}x_1 \\ \vdots \\ a_{1m}x_1 \end{bmatrix} + \begin{bmatrix} a_{21}x_2 \\ a_{21}x_2 \\ \vdots \\ a_{2m}x_1 \end{bmatrix} + \dots + \begin{bmatrix} a_{n1}x_n \\ a_{n2}x_n \\ \vdots \\ a_{nm}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector scaling

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector addition

General Vector Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

by vector equality

The Fundamental Connection

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

The Fundamental Connection

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vector equation

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system of linear equations

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vector equation

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augmented matrix

*This is our big
perspective shift
of the day*

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

How To: Linear Combination Problems

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this is notation for
building a matrix
out of column
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Question

Can $\begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ be written as a linear combination of $\begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$?

Answer

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Df. the *span* of a set of vectors is the set of all possible linear combinations of them

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read: $\mathbf{u}$ is an element of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$



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demo
(from ILA)

Span of One Vector

Span of Two Vector

Spans (Geometrically)

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for one vector

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this is **all scalar multiple of $\mathbf{v}$**

the span of one vector is a **line**

Spans (Geometrically)

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the span of **two** vectors can be a **plane**

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!!IMPORTANT!!

In all cases they pass through the origin

demo
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How To: Span Problems

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you know how to do this now

Example

$$\text{Is } \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \text{ in span } \left\{ \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} \right\} ?$$

Question (Conceptual)

What does it mean geometrically if $\mathbf{b} \notin \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

demo
(from ILA)

How To: Inconsistency and Spans

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There is **no way** to write $\mathbf{b}$ as a linear combination

Example

*Find a vector **not** in* $\text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 3 \end{bmatrix} \right\}$

Summary

Vectors are fundamental objects

We can think of them as the **columns** of a linear system

We can **scale** them and **add** them together

They can **span** spaces which represent **hyperplanes**